

# Learning to share sunlight: differentiable-simulation design and control of climate-resilient agrivoltaics across the world's staple croplands

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## Abstract

Solar power and food production increasingly compete for the same land, while warming exposes crops to more heat and drought. Agrivoltaic systems, which grow crops under photovoltaic (PV) modules, can relieve this conflict, but how much electricity farmland can produce without compromising food security depends on how sunlight is shared between modules and plants, day by day, for every crop and climate. We propose SALS (stress-aware light sharing), an algorithm that learns the design and the daily control of single-axis tracker agrivoltaics end-to-end through a differentiable simulator. The simulator couples solar geometry, backtracking trackers, PV output, crop-level light and microclimate, a soil-water balance and the SIMPLE crop model with heat and drought stress. A design network maps site descriptors to the ground coverage ratio, and a control network maps the daily crop, soil and weather state to the tracker's light-sharing level. Both are trained by back-propagation through whole seasons to maximise electricity subject to a 90% yield-retention floor. We applied SALS to 400 sites representing the global harvested area of wheat, rice, maize, soybean and potato, with NASA POWER weather for 2001–2019 and warming of +1.5, +2 and +3 °C. On geographically held-out sites and years, SALS produced 1019 MWh ha<sup>-1</sup> yr<sup>-1</sup> (72% of a conventional PV plant) at a mean yield retention of 0.91. This gave a land-equivalent ratio of 1.63, compared with 1.20 for static agrivoltaic designs and 1.42 for an expert stress rule ( $p < 10^{-4}$ ). Its advantage grew with warming, to an LER of 1.75 at +3 °C, as the learned policy shifted from sharing light during canopy development to shading crops during heat. The yield floor was met on average but in only 45–49% of individual seasons, which calls for chance-constrained extensions. Scaled to 5% of the harvested

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area of the five crops, learned agrivoltaics has a technical potential of about 31,000 TWh yr<sup>-1</sup>, comparable to current global electricity generation.

7 *Keywords:* Agrivoltaics, Differentiable simulation, Food–energy nexus,  
8 Climate adaptation, Solar tracking, Crop modelling, Physics-informed  
9 machine learning

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## 10 1. Introduction

11 Two of the defining challenges of this century compete for the same re-  
12 source: land. The energy transition requires terawatts of new solar pho-  
13 tovoltaic (PV) capacity, and ground-mounted solar farms need roughly one  
14 to two hectares per megawatt. Food production must rise while crops are  
15 exposed to more frequent heat waves and droughts. Warming alone is pro-  
16 jected to reduce global yields of maize, wheat, rice and soybean by several  
17 per cent per degree (Zhao et al., 2017), and a new generation of climate and  
18 crop models shows these impacts emerging earlier than previously expected  
19 (Jägermeyr et al., 2021). Croplands are also where solar power potential is  
20 greatest (Adeh et al., 2019). They are flat, sunny, accessible and close to  
21 rural electricity demand, so converting farmland into solar farms is tempting  
22 and controversial at the same time.

23 Agrivoltaic (AV) systems combine crops and PV modules on the same  
24 land and aim to resolve this conflict (Dupraz et al., 2011; Dinesh and Pearce,  
25 2016; Weselek et al., 2019). By sharing sunlight between modules and plants,  
26 a hectare can deliver more food plus energy than the two uses side by side,  
27 with land-equivalent ratios (LER) well above one in field trials and mod-  
28 elling studies (Dupraz et al., 2011; Amaducci et al., 2018; Valle et al., 2017;  
29 Trommsdorff et al., 2021). The partial shade also changes the crop micro-  
30 climate. It lowers canopy temperature and evaporative demand, conserves  
31 soil moisture and improves water-use efficiency, especially in drylands (Mar-  
32 rou et al., 2013; Hassanpour Adeh et al., 2018; Barron-Gafford et al., 2019;  
33 Elamri et al., 2018; Weselek et al., 2021). Agrivoltaics is therefore a candi-  
34 date adaptation measure against heat and drought stress, in addition to a  
35 mitigation measure through renewable electricity. The same shade, however,  
36 also reduces the light available for photosynthesis. Regulators increasingly  
37 require AV systems to retain most of the reference crop yield. The German  
38 pre-standard DIN SPEC 91434, for example, requires at least 66% of the  
39 reference yield (DIN, 2021), and other schemes and many farmers set stricter

40 floors. How much electricity can be produced without sacrificing food secu-  
41 rity depends on the crop, the local climate, the year and the growth stage.

42 Most AV designs fix the geometry of the array, i.e. the ground coverage  
43 ratio (GCR, module area per ground area) and the tilt or tracking rule, from  
44 generic rules of thumb. Single-axis trackers add a degree of freedom: the  
45 modules can be rotated away from the sun to let light through when the crop  
46 needs it, as demonstrated for lettuce by Valle et al. (2017), or towards the  
47 sun to maximise electricity and shade the crop during heat waves. Choosing  
48 the rotation every day is a sequential decision problem. The value of light  
49 depends on the crop’s phenological stage, soil water and upcoming weather,  
50 and the effect of a decision appears in the harvest months later. Designing  
51 the GCR and the control policy for every combination of crop and climate  
52 on Earth, and for a warming climate, is beyond manual rule design. Recent  
53 work in this journal has used evolutionary multi-objective optimisation to  
54 lay out PV arrays on farmland (Zhang et al., 2026). Machine learning for  
55 crop management has mainly used reinforcement learning with black-box  
56 simulators (Gautron et al., 2022), which is sample-inefficient and hard to  
57 scale to global, multi-decadal assessments.

58 This paper proposes *SALS* (stress-aware light sharing), a learning algo-  
59 rithm that designs and controls agrivoltaic systems jointly, for any cropland  
60 site on Earth, through a differentiable physics simulator. The simulator  
61 couples hourly solar geometry, single-axis trackers with backtracking, row-  
62 to-row shading and PV electricity, the direct and diffuse light reaching the  
63 crop, a microclimate model of canopy temperature and reference evapotran-  
64 spiration under the modules, and the SIMPLE crop model with heat and  
65 drought stress (Zhao et al., 2019). Because every component is written as  
66 differentiable tensor operations, two small neural networks can be trained  
67 by gradient descent through entire growing seasons. A *design network* maps  
68 site descriptors to the GCR, and a *control network* maps the daily crop and  
69 soil state and the same-day weather forecast to the tracker’s light-sharing  
70 level. Training maximises electricity subject to a yield-retention floor, across  
71 hundreds of sites, twenty years of weather and four warming levels. This  
72 physics-informed learning approach (Karniadakis et al., 2021) gives a single  
73 policy that generalises to unseen regions, years and climates.

74 We apply SALS to 400 sites that represent where maize, wheat, rice,  
75 soybean and potato are grown worldwide, sampled in proportion to their  
76 harvested area (Yu et al., 2020) with local crop calendars (Jägermeyr et al.,  
77 2021), and driven by NASA POWER daily weather for 2001–2019 (Sparks,

2018). The contributions are as follows.

1. A differentiable agrivoltaic simulator that couples tracker geometry, PV output, crop-level light and microclimate, soil water and a generic crop model, and runs thousands of site-seasons in parallel on a CPU.
2. The SALS algorithm, which learns the array design and a closed-loop, stress-aware tracker policy end-to-end under a food-security constraint. It is evaluated on geographically held-out sites and years against static designs, expert rules and a perfect-foresight oracle.
3. A global assessment under warming of +1.5, +2 and +3 °C, which quantifies how much electricity croplands can supply while retaining at least 90% of crop yields, and where agrivoltaics protects yields against heat and drought.

## 2. Materials and methods

### 2.1. Global cropland sites, weather and crop calendars

We studied five staple crops that together cover about 670 Mha of harvested area: wheat, rice, maize, soybean and potato. Harvested areas (rain-fed and irrigated) were taken from the SPAM 2010 gridded production maps (Yu et al., 2020) and aggregated to 1° cells. For each crop, 80 cells with at least 5000 ha of the crop were drawn without replacement with probability proportional to harvested area. An unweighted average over the sample therefore approximates an area-weighted average over global production. The 400 crop–site pairs lie in 381 distinct cells in 50 countries (Fig. 2). Sowing and maturity dates came from the GGCM Phase 3 crop calendar at 0.5° (Jägermeyr et al., 2021). For wheat we used the winter-wheat calendar unless the coldest-month mean temperature was below  $-8^{\circ}\text{C}$ , in which case the spring-wheat calendar was used. Daily weather for 2001–2020 was retrieved from NASA POWER (Sparks, 2018): global and diffuse shortwave irradiance, maximum, minimum and dew-point temperature, precipitation and wind speed at 2 m. Each site contributes 19 seasons (sowing years 2001–2019), each simulated as the 365 days after sowing, which gives 7600 site-seasons.

Climate change was represented by the global warming levels +1.5, +2 and +3 °C relative to 2001–2020. Daily maximum, minimum and dew-point temperatures were shifted uniformly, which keeps relative humidity approximately constant and raises the vapour-pressure deficit. The atmospheric

CO<sub>2</sub> concentration was set to 390, 450, 500 and 600 ppm for the baseline and the three warming levels. Precipitation and irradiance were left unchanged. This delta approach isolates the thermal signal of warming; its limitations are discussed in Section 4.3.

## 2.2. A differentiable agrivoltaic simulator

The simulator advances all site-seasons in parallel with a daily time step for the crop and an hourly time step for radiation and PV (Fig. 1). All equations are implemented as PyTorch tensor operations (Paszke et al., 2019), so that yields and electricity are differentiable with respect to the design and control variables.

*Hourly irradiance.* Daily global and diffuse irradiance were disaggregated into hourly values with the Collares-Pereira and Rabl (1979) and Liu and Jordan (1960) ratios and renormalised to the daily totals. Beam irradiance on the horizontal is the difference between the two. Solar position follows the standard declination and hour-angle equations.

*Array geometry and PV electricity.* We consider elevated horizontal single-axis trackers with north–south axes. The axis height is 1.25 module widths, which leaves space for machinery under the modules. The rotation  $\phi$  is limited to  $\pm 60^\circ$ , and the ground coverage ratio  $g$  is the module width divided by the row pitch. Let  $\psi$  be the solar zenith angle projected onto the east–west vertical plane. The energy-optimal rotation with backtracking (Lorenzo et al., 2011) is  $\phi_{\text{bt}} = \psi - \text{sgn}(\psi) \arccos(\cos \psi / g)$  when  $\cos \psi < g$ , and  $\phi_{\text{bt}} = \psi$  otherwise. The *light-sharing* rotation  $\phi_{\text{sh}}$  is the feasible angle closest to edge-on to the sun, which minimises the shadow on the crop. Given a light-sharing level  $u \in [0, 1]$  chosen once per day, the rotation in every hour is

$$\phi = (1 - u) \phi_{\text{bt}} + u \phi_{\text{sh}}. \quad (1)$$

Plane-of-array irradiance combines beam irradiance on the module (reduced by the fraction of the module shaded by the neighbouring row,  $\max\{0, 1 - \cos \psi / (g |\cos(\phi - \psi)|)\}$ ), isotropic sky diffuse and ground-reflected light (albedo 0.2). The cell temperature follows the NOCT model (45 °C), with an hourly air temperature interpolated sinusoidally between the daily extremes. Electricity per unit land is plane-of-array irradiance  $\times \eta \times g$ , with  $\eta = 0.21 [1 - 0.0035(T_{\text{cell}} - 25)] (1 - 0.14)$  for module efficiency, temperature losses and system losses. A conventional ground-mounted PV plant ( $g = 0.4$ , backtracking, no crop) is the energy reference  $E_{\text{PV}}$ .

147 *Light and microclimate at the crop.* Averaged over the row pitch, the fraction  
 148 of ground reached by beam light is  $1 - \min\{1, g |\cos(\phi - \psi)| / \cos \psi\}$ . The  
 149 fraction of sky diffuse reaching the ground is a two-dimensional isotropic sky-  
 150 view factor. We pre-computed it for 25 values of  $g$  and 13 rotation angles by  
 151 ray casting over 31 neighbouring rows, and interpolate it bilinearly (differenti-  
 152 ably). The daily ratio  $s$  of the irradiance at the crop to that of the open field  
 153 defines the crop’s shading. Following observations of cooler canopies under  
 154 PV modules (Marrou et al., 2013; Barron-Gafford et al., 2019), the canopy  
 155 maximum temperature is lowered by  $\kappa(1 - s)$  with  $\kappa = 2^\circ\text{C}$  (a sensitivity  
 156 analysis over  $\kappa$  is reported in Section 3.5). The reference evapotranspiration  
 157 under the modules,  $\text{ET}_0$ , is recomputed with the FAO-56 Penman–Monteith  
 158 equation (Allen et al., 1998) from the reduced radiation and canopy temper-  
 159 ature. Soil water conservation under the modules (Hassanpour Adeg et al.,  
 160 2018; Elamri et al., 2018) therefore emerges from the model and is not im-  
 161 posed.

162 *Crop growth.* Crop growth follows the SIMPLE model (Zhao et al., 2019)  
 163 with the species parameters of its Table 1a (Table 1). Daily biomass growth  
 164 is

$$\Delta B = R_{\text{crop}} \cdot \text{RUE} \cdot f_{\text{Solar}} \cdot f_{\text{CO}_2} \cdot f_{\text{Temp}} \cdot \min\{f_{\text{Heat}}, f_{\text{Water}}\}, \quad (2)$$

165 where  $R_{\text{crop}} = s R$  is the radiation received by the crop.  $f_{\text{Solar}}$  is the canopy’s  
 166 radiation interception, driven by thermal time and accelerated senescence  
 167 under heat and drought stress.  $f_{\text{Heat}}$  decreases linearly between the heat  
 168 threshold  $T_{\text{heat}}$  and the extreme temperature  $T_{\text{ext}}$  of the canopy maximum  
 169 temperature.  $f_{\text{Water}} = 1 - S_{\text{water}} \text{ARID}$ , with the agricultural reference index  
 170 for drought (Woli et al., 2012) computed from a daily soil-water balance with  
 171 curve-number runoff, drainage and root water uptake. Irrigated fields have  
 172  $f_{\text{Water}} = 1$ . Yields of the rainfed and irrigated parts of a site are combined  
 173 with the SPAM irrigated fraction. The thermal-time requirement to maturity  
 174 was calibrated for every site as the mean thermal time between the GGCMI  
 175 sowing and maturity dates under the 2001–2020 climate, and the canopy  
 176 parameters were scaled accordingly. The cultivar is thus adapted to the  
 177 present climate, and warming shortens the season. Growth stops smoothly  
 178 once the thermal-time requirement is met, and yield is biomass multiplied  
 179 by the harvest index. Because SIMPLE uses a linear radiation-use efficiency,  
 180 it does not credit the higher light-use efficiency of shaded canopies. Shading  
 181 losses are therefore, if anything, overestimated.

182 *2.3. SALS: stress-aware light sharing*

183 SALS learns two functions (Algorithm 1):

- 184 • a *design network*  $g = D_\theta(\mathbf{z})$ , a multilayer perceptron with two hidden  
 185 layers of 32 units, which maps site descriptors  $\mathbf{z}$  to a ground coverage  
 186 ratio in  $[0.05, 0.60]$ . The descriptors are the crop (one-hot), latitude,  
 187 mean irradiance, growing-season maximum temperature, precipitation  
 188 and vapour-pressure deficit, the frequency of heat-stress days and the  
 189 irrigated fraction;
- 190 • a *control network*  $u_t = \pi_\omega(\mathbf{x}_t)$ , a multilayer perceptron with two hidden  
 191 layers of 64 units and a sigmoid output, which chooses the daily light-  
 192 sharing level of Eq. (1) from information available on day  $t$ : relative  
 193 thermal time  $TT/T_{\text{sum}}$ , radiation interception  $f_{\text{Solar}}$ , relative soil water,  
 194 the previous day’s ARID, the day’s forecast of maximum temperature  
 195 relative to  $T_{\text{heat}}$  and  $T_{\text{opt}}$ , irradiance and  $ET_0$ , the irrigation status, the  
 196 design  $g$ , latitude, day of season and the crop.

197 Outside the cropping season the trackers backtrack for maximum electricity.  
 198 For site-season  $i$  with open-field yield  $Y_i^{\text{open}}$ , let  $r_i = Y_i/Y_i^{\text{open}}$  be the yield  
 199 retention and  $e_i = E_i/E_{\text{PV},i}$  the electricity relative to a conventional PV  
 200 plant. SALS solves

$$\max_{\theta, \omega} \mathbb{E}_i[e_i] \quad \text{s.t.} \quad r_i \geq \rho \quad \forall i, \quad (3)$$

201 where  $\rho$  is the food-security floor ( $\rho = 0.9$  unless stated otherwise). It does  
 202 so with the penalty objective

$$\mathcal{L}(\theta, \omega) = \mathbb{E}_{i \sim \mathcal{B}} \left[ -e_i + \mu \max\{0, \rho - r_i\}^2 \right], \quad \mu = 30. \quad (4)$$

203 Because  $r_i$  and  $e_i$  are outputs of the differentiable simulator, the exact gradi-  
 204 ent of Eq. (4) with respect to  $(\theta, \omega)$  is obtained by back-propagation through  
 205 the 365 daily steps of every season (back-propagation through time). This  
 206 is the key difference from reinforcement learning with black-box simulators  
 207 (Gautron et al., 2022), which estimates gradients from sampled returns. Each  
 208 minibatch contains 512 site-seasons, each duplicated into a rainfed and an  
 209 irrigated field, drawn from one of the four climates in turn. The networks  
 210 are trained with Adam (Kingma and Ba, 2015) (learning rate  $3 \times 10^{-3}$ , co-  
 211 sine decay, gradient-norm clipping at 1) for 400 steps, i.e. about 50 passes

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**Algorithm 1** SALS training

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**Require:** training site-seasons in four climates; floor  $\rho$ ; penalty  $\mu$

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1: initialise design network  $D_\theta$  and control network  $\pi_\omega$ 
2: for step =  $1, \dots, N$  do
3:   sample a climate and a minibatch  $\mathcal{B}$  of site-seasons (rainfed and irrigated rows)
4:    $g_i \leftarrow D_\theta(\mathbf{z}_i)$ ; initialise crop and soil states
5:   for day  $t = 1, \dots, 365$  do
6:      $u_{it} \leftarrow \pi_\omega(\mathbf{x}_{it})$  if the crop is in the field, else 0
7:     hourly rotation (Eq. 1), PV electricity, crop light, canopy temperature,  $ET_0$ 
8:     advance the soil-water balance and the SIMPLE crop state (Eq. 2)
9:      $r_i \leftarrow Y_i/Y_i^{\text{open}}$ ,  $e_i \leftarrow E_i/E_{\text{PV},i}$ ; loss  $\mathcal{L}$  (Eq. 4)
10:    back-propagate through the season; Adam update of  $(\theta, \omega)$ 
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212 over the training seasons. Growth-stage logistic functions are evaluated in a  
213 numerically stable form, so that gradients remain finite even under extreme  
214 heat.

#### 215 2.4. Baselines

216 We compared SALS with the following references and baselines.

- 217 • *Open field*: no modules ( $r = 1$ ,  $e = 0$ , LER = 1).
- 218 • *PV plant*: conventional ground-mounted PV without a crop ( $e = 1$ ,  
219 LER = 1).
- 220 • *AV static (rule)*: backtracking trackers with one GCR per crop, the  
221 largest value on a grid  $\{0.05, 0.10, \dots, 0.50\}$  whose mean retention on  
222 the training seasons reaches  $\rho$ . This mimics a design guideline.
- 223 • *AV static (oracle GCR)*: backtracking trackers with the largest GCR  
224 that meets  $\rho$  on average over each test site’s own seasons. The GCR is  
225 chosen with knowledge of the test weather.
- 226 • *Seasonal sharing*: the light-sharing rotation ( $u = 1$ ) throughout the  
227 cropping season, with the per-site oracle GCR.

Table 1: SIMPLE crop parameters (Zhao et al., 2019, Table 1a).  $T_{\text{sum}}$ ,  $I_{50A}$  and  $I_{50B}$  are rescaled per site to the local crop calendar (Section 2.2). Units: temperatures in °C, thermal times in °Cd, RUE in gMJ<sup>-1</sup>;  $S_{\text{CO}_2}$  is the relative RUE increase per 100 ppm above 350 ppm.

| Crop (cultivar)       | $T_{\text{sum}}$ | HI   | $I_{50A}$ | $I_{50B}$ | $T_{\text{base}}$ | $T_{\text{opt}}$ | RUE  | $I_{50 \text{ max } H}$ | $I_{50 \text{ max } W}$ | $T_{\text{heat}}$ | $T_{\text{ext}}$ | $S_{\text{CO}_2}$ | $S_{\text{water}}$ |
|-----------------------|------------------|------|-----------|-----------|-------------------|------------------|------|-------------------------|-------------------------|-------------------|------------------|-------------------|--------------------|
| Wheat (Batten)        | 2150             | 0.34 | 280       | 50        | 0                 | 15               | 1.24 | 100                     | 25                      | 34                | 45               | 0.08              | 0.4                |
| Rice (IR72)           | 2300             | 0.47 | 850       | 200       | 9                 | 26               | 1.24 | 100                     | 10                      | 34                | 50               | 0.08              | 1.0                |
| Maize (McCurdy 84aa)  | 2050             | 0.50 | 500       | 50        | 8                 | 28               | 2.10 | 100                     | 12                      | 34                | 50               | 0.01              | 1.2                |
| Soybean (Williams 82) | 2350             | 0.40 | 600       | 200       | 6                 | 27               | 0.86 | 120                     | 20                      | 36                | 50               | 0.07              | 0.9                |
| Potato (Sebago)       | 2400             | 0.85 | 500       | 350       | 4                 | 22               | 1.30 | 50                      | 30                      | 34                | 45               | 0.10              | 0.4                |

- *Stress rule*: an expert rule that shares light ( $u = 1$ ) during the main growth phase ( $0.1 < \text{TT}/T_{\text{sum}} < 0.9$ ) except on days with a heat-stress forecast ( $T_{\text{max}} \geq T_{\text{heat}}$ ) or drought ( $\text{ARID} > 0.5$ ), when the modules track the sun and shade the crop. It uses the per-site oracle GCR.
- *Open-loop optimisation (perfect foresight)*: the GCR and all 365 daily controls of every test season are optimised directly by gradient descent through the simulator (150 Adam steps, learning rate 0.1) with perfect knowledge of that season’s weather. It is a per-season trajectory optimisation that no deployable controller could implement. Because the problem is non-convex and each season is optimised in isolation, it is not guaranteed to be an upper bound.

The three oracle-GCR baselines and the open-loop optimisation therefore have access to the test weather, whereas SALS chooses its design from climate descriptors alone.

## 2.5. Evaluation protocol and metrics

Sites were split spatially:  $10^\circ \times 10^\circ$  blocks were randomly assigned to training (70%) or test (30%), so that no test site has a training site in the same block. Seasons sown in 2001–2014 at training sites were used for training, and seasons sown in 2015–2019 at test sites for evaluation, so that the test set is out of sample in both space and time. We report the mean electricity (MWh ha<sup>-1</sup> yr<sup>-1</sup>) and relative electricity  $e$ , the mean yield retention  $r$ , the *compliance* (share of test seasons with  $r \geq \rho$ ), and the land-equivalent ratio  $\text{LER} = r + e$  (Dupraz et al., 2011). Differences between methods were tested with paired Wilcoxon signed-rank tests over test seasons, and uncertainty is given as bootstrap 95% confidence intervals over sites.

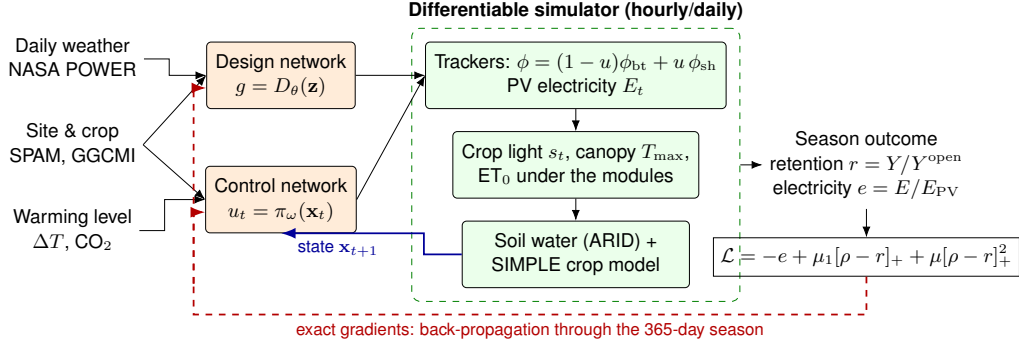


Figure 1: SALS. A design network chooses the ground coverage ratio  $g$  of single-axis tracker rows from site descriptors, and a control network chooses the daily light-sharing level  $u_t$  from the crop, soil and weather state. The differentiable simulator converts these decisions into PV electricity, light and microclimate at the crop, soil water and crop growth. The season’s electricity and yield retention enter a penalised objective whose exact gradient is back-propagated through the 365-day season to both networks.

### 3. Results

#### 3.1. Electricity and food on held-out sites and years

Table 2 and Fig. 3 compare the methods on 461 test seasons: 2015–2019 at geographically held-out sites, excluding seasons in which the open-field crop failed ( $< 0.2 \text{ t ha}^{-1}$ ). Under the baseline climate, the static agrivoltaic designs that keep 90% of the yield must use very sparse arrays (GCR  $\approx 0.10$ ). They produce 361–377 MWh  $\text{ha}^{-1} \text{ yr}^{-1}$ , a quarter of the electricity of a conventional PV plant, and reach an LER of 1.18–1.20. Rotating the trackers to share light with the crop allows denser arrays. The seasonal-sharing and stress rules reach 629 and 707 MWh  $\text{ha}^{-1} \text{ yr}^{-1}$  (LER 1.37 and 1.42), even though their GCR is chosen with knowledge of the test weather. SALS chooses a GCR of 0.34 on average and produces 1019 MWh  $\text{ha}^{-1} \text{ yr}^{-1}$ , 72% of a conventional PV plant, with a mean yield retention of 0.909. Its LER of 1.63 (95% CI 1.57–1.70) is 0.21 higher than that of the best expert rule and 0.43 higher than that of the static design ( $p < 10^{-4}$ , paired Wilcoxon tests, Table A.3). Per-season open-loop optimisation with perfect foresight reached an LER of 1.55. It remained below SALS because optimising 365 daily controls separately for each season is a harder, non-convex problem than learning one policy from thousands of seasons.

The gain comes with a trade-off that must be stated clearly. SALS meets

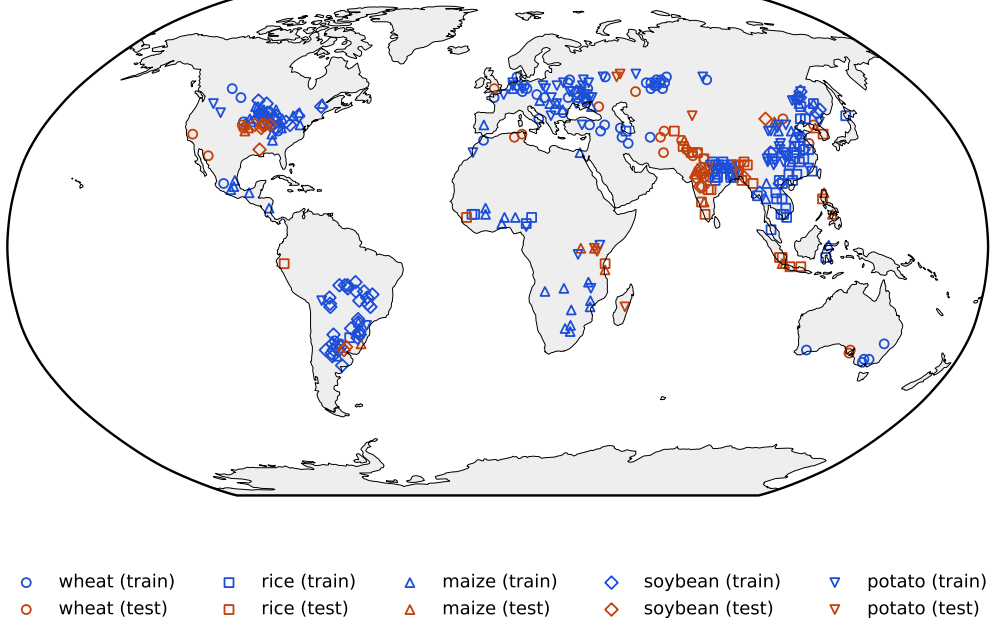


Figure 2: The 400 crop–site pairs, sampled in proportion to the harvested area of each crop (SPAM 2010). Blue: training sites; orange: geographically held-out test sites (10° blocks).

273 the floor *on average*: its mean retention is 0.90–0.91 in all four climates. It  
 274 meets the floor in only 45–49% of individual seasons, however, whereas the  
 275 rule-based baselines meet it in 78–85% of seasons. The penalty objective  
 276 (Eq. 4) lets SALS go slightly below the floor in some seasons in exchange for  
 277 much more electricity. The rules stay above the floor with a margin (mean  
 278  $r \approx 0.92$ ) and produce 30–65% less electricity. The floor in Eq. (3) is a per-  
 279 season constraint; a formulation that guarantees it in a given share of seasons,  
 280 i.e. a chance constraint, would be the natural extension for deployments  
 281 where regulation is enforced every season (Section 4.3).

### 282 3.2. What the policy learned

283 Fig. 4 shows the learned control for two of the hottest test seasons under  
 284 +3 °C. For rainfed maize in the Nile Delta, SALS keeps the trackers in energy  
 285 mode during crop establishment, opens them fully ( $u \approx 1$ ) during the canopy-  
 286 building phase, when intercepted light is most valuable, and returns to energy  
 287 mode in the second half of the season, when maximum temperatures exceed

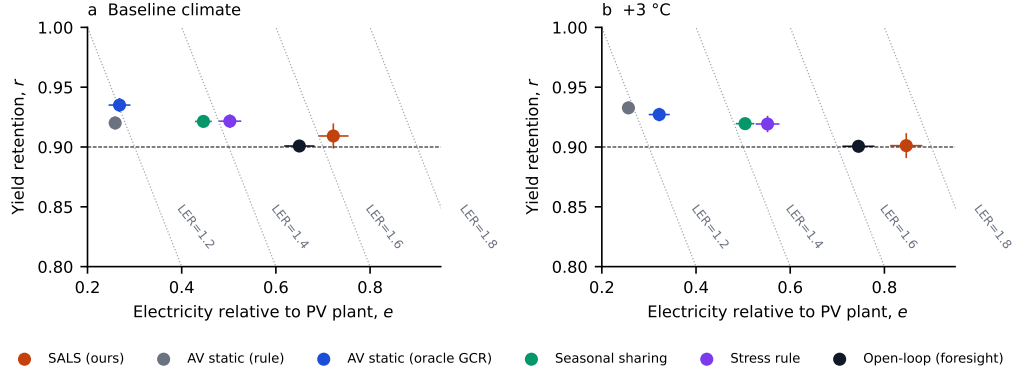


Figure 3: Electricity–food trade-off on held-out sites and years. Points are means with 95% confidence intervals; dotted lines are iso-LER lines; the dashed line is the 90% yield-retention floor.

288 40 °C and the shade protects the canopy. For irrigated wheat in north-western  
 289 India, it shares light throughout the cool vegetative phase and shades the crop  
 290 during the hot grain-filling period before harvest. Over all active crop days  
 291 of the +3 °C test seasons, the mean light-sharing level is 0.33 on days with  
 292 heat stress and 0.46 on other days. The policy has thus learned to trade light  
 293 for cooling when heat, not light, limits growth. The design network chooses  
 294 denser arrays for hotter and drier sites and for maize and soybean (mean  
 295 GCR 0.42 and 0.39 at present climate) than for wheat and potato (0.32). In  
 296 these crops, heat and drought limit yield more than light does.

### 297 3.3. Agrivoltaics under global warming

298 Warming changes the balance between light and heat. With the same  
 299 trained networks, SALS increases its electricity from 1019 to 1079, 1106  
 300 and 1173 MWh ha<sup>-1</sup> yr<sup>-1</sup> at +1.5, +2 and +3 °C (from 72% to 85% of a  
 301 conventional PV plant) while keeping the mean retention relative to the  
 302 same-climate open field at 0.90 (Table 2). Its LER rises from 1.63 to 1.75,  
 303 whereas that of the static design stays at about 1.2. Two effects cause this.  
 304 Warmer seasons are shorter, so the crop occupies the field for fewer days,  
 305 and heat and drought stress make light less limiting, so more shade can be  
 306 tolerated or is even beneficial. Across all 400 sites, the SALS system yields  
 307 more than the open field in 13–14% of seasons, rising to 31–35% for maize  
 308 and 25–28% for soybean, mostly at hot, rainfed sites (Fig. 6b).

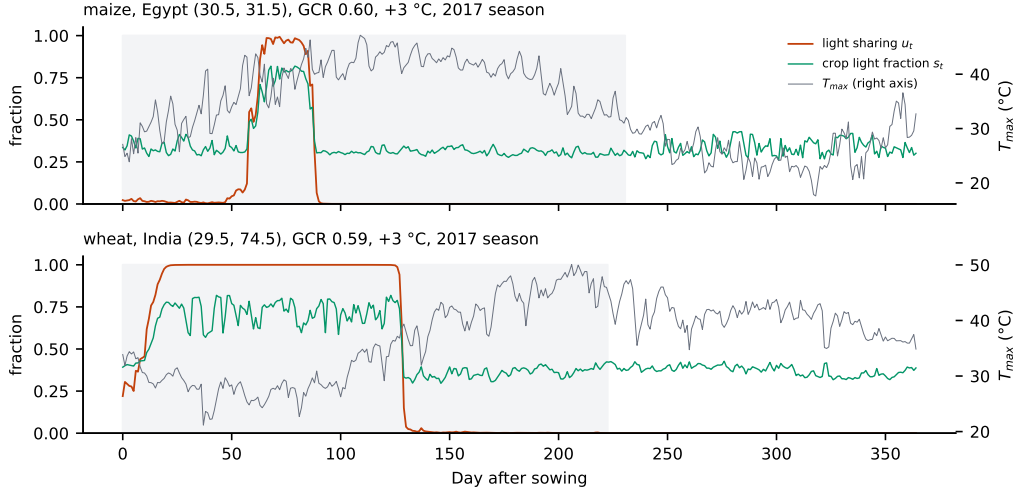


Figure 4: Learned daily control for two hot test seasons under  $+3\text{ }^{\circ}\text{C}$ . Shading marks the cropping season. Orange: light-sharing level  $u_t$  chosen by SALS; green: fraction of open-field irradiance reaching the crop; grey: daily maximum temperature (right axis).

309 Agrivoltaics is not, however, a remedy for the aggregate warming damage.  
 310 Summed over sites, open-field production of maize and rice falls by 18 and  
 311 17% at  $+3\text{ }^{\circ}\text{C}$  in our simulations, and SALS production by 23 and 26% relative  
 312 to the baseline open field, i.e. 94 and 90% of the same-climate open field  
 313 (Fig. 5). Potato and wheat, which benefit from warmer conditions at cool  
 314 sites and from  $\text{CO}_2$  fertilisation in the model, gain in the open field, and  
 315 SALS keeps 87–89% of this production. The benefit of SALS under warming  
 316 is that it maintains the food-security floor while producing more electricity  
 317 per hectare. It does not reverse the climate impact on yields.

### 318 3.4. Global electricity potential

319 Fig. 6 maps the electricity and LER of SALS at all 400 sites. Electricity  
 320 ranges from a few hundred  $\text{ha}^{-1}\text{yr}^{-1}$  at high-latitude wheat sites to more than  
 321 2000  $\text{MWh ha}^{-1}\text{yr}^{-1}$  at sunny, hot maize and soybean sites in South Asia,  
 322 Africa and the Americas, where the LER exceeds 2. Because the sites were  
 323 sampled in proportion to harvested area, their mean electricity per hectare  
 324 can be scaled to the global area of each crop. Deploying SALS agrivoltaics  
 325 on 5% of the harvested area of the five crops (33 Mha) would generate about  
 326 31,000  $\text{TWh yr}^{-1}$  at present climate and 37,000  $\text{TWh yr}^{-1}$  at  $+3\text{ }^{\circ}\text{C}$ , while  
 327 retaining on average 93% of the crop yield on that land. Static agrivoltaic

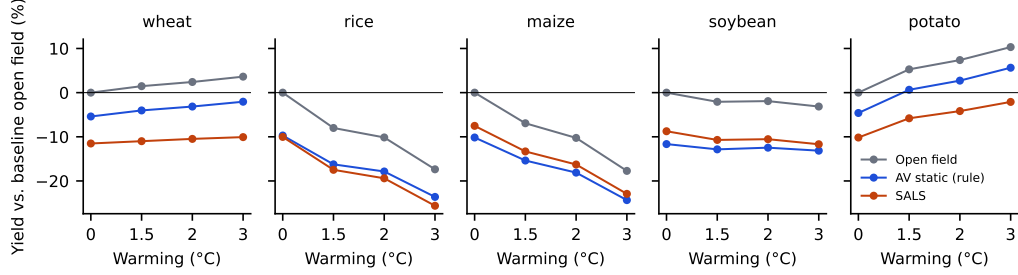


Figure 5: Production relative to the baseline open field (sum over sites, test years) as a function of warming, for the open field, static agrivoltaics and SALS.

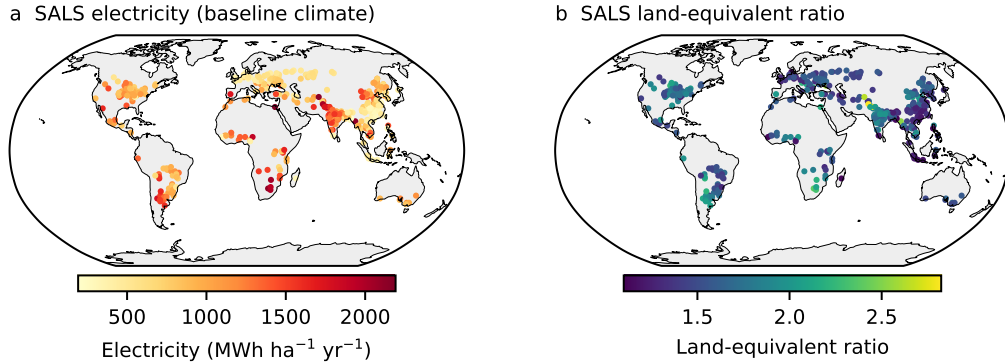


Figure 6: SALS electricity (a) and land-equivalent ratio (b) at the 400 sites (baseline climate, mean of seasons 2015–2019).

328 designs with the same yield floor would generate about  $11,700 \text{ TWh yr}^{-1}$ .  
 329 For comparison, global electricity generation was about  $30,000 \text{ TWh}$  in 2023  
 330 (Ember, 2024). These are technical potentials: they ignore grid integration,  
 331 storage, land tenure and economic constraints (Section 4.3). They show,  
 332 however, that light sharing makes the food-compatible share of cropland  
 333 large enough to matter for global energy security.

### 334 3.5. Sensitivity and ablation

335 The canopy-cooling coefficient  $\kappa$  is the most uncertain microclimate pa-  
 336 rameter. Evaluated without retraining at  $\kappa = 0, 1, 2$  and  $3^\circ\text{C}$ , the electricity  
 337 of SALS barely changes ( $1026\text{--}1022 \text{ MWh ha}^{-1} \text{yr}^{-1}$  at present climate). Its  
 338 mean retention changes from  $0.871$  to  $0.931$  at present climate and from  
 339  $0.847$  to  $0.927$  at  $+3^\circ\text{C}$ . Without any cooling benefit, the policy therefore

under-shoots the floor by 3–5 points. The heat-protection effect is thus an important part of the benefit under warming, and field measurements of canopy temperature under trackers are needed to pin it down. To isolate the contribution of the stress-aware inputs, we trained a variant whose control network sees only phenology, time, design and crop, with soil water, ARID, the maximum-temperature, irradiance and  $ET_0$  forecasts masked. Both variants were trained for the same budget of 200 steps and evaluated on the held-out seasons. At similar mean retention (0.898 vs 0.897 at present climate), removing the stress inputs reduced the relative electricity from 0.69 to 0.50 and the LER from 1.59 to 1.40. Under  $+3^\circ\text{C}$  the drop was from 1.67 to 1.42. The stress-blind policy performs like the expert rules: it can learn *when* in the season to share light, but not to respond to the heat and drought of a particular season. Training longer (400 vs 200 steps) added a further 0.04–0.08 to the LER.

## 4. Discussion

### 4.1. *Why learning the control matters*

The results show that most of the electricity that agrivoltaics can add without compromising food production comes from *when* light is shared rather than from the array geometry alone. With static trackers, the yield floor forces sparse arrays that cover a tenth of the ground. Rules that share light during the season double the electricity, and a learned, state-dependent schedule nearly triples it. SALS concentrates the crop’s light in the phases where intercepted radiation is converted into biomass most efficiently (canopy build-up) and when light, rather than heat or water, limits growth. In the remaining days it produces electricity and shades the crop against heat. This agrees with field observations that shading is less damaging, or even beneficial, when it coincides with heat or drought (Marrou et al., 2013; Barron-Gafford et al., 2019; Weselek et al., 2021), and with the dynamic-tracking experiments of Valle et al. (2017). SALS generalises this idea across crops and climates through a policy learned from data rather than from hand-crafted rules.

Differentiable simulation is what makes this learning tractable. A season has 365 coupled decisions and a single delayed outcome, the harvest. Exact gradients through the physics give credit to each daily decision without the variance of sampled returns that makes reinforcement learning with black-box crop simulators expensive (Gautron et al., 2022). Training one policy on

about 4000 seasons in four climates took about 40 minutes on four CPU cores, and the policy transfers to unseen regions and years. Interestingly, the policy outperformed per-season trajectory optimisation with perfect foresight. Sharing one policy across thousands of seasons regularises the non-convex control problem, a benefit of amortised optimisation.

#### 4.2. *Implications for energy security and climate adaptation*

Agrivoltaics has been advocated as a way to decarbonise electricity without taking land from food (Dinesh and Pearce, 2016; Adeh et al., 2019). Our results quantify this potential under an explicit food-security floor and show how strongly it depends on control. With static designs, 5% of the area of five staple crops would supply about 40% of today’s global electricity. With learned light sharing it would supply roughly all of it, on land that keeps producing 93% of its food. The potential grows with warming because heat makes shade less costly. Agrivoltaics therefore becomes more attractive as the climate warms, in the regions of South Asia, Africa and the Americas where both electricity demand and heat risks for crops are rising. SALS does not, however, reverse the negative impact of warming on the aggregate yields of maize and rice. Agrivoltaics complements breeding, irrigation and management adaptation; it does not replace them.

#### 4.3. *Limitations*

Several simplifications bound the conclusions. (i) The crop model is generic. SIMPLE has a linear radiation-use efficiency, so it does not reward the higher light-use efficiency of shaded canopies, and it represents heat stress through maximum temperature only. Crop-specific shade responses (Marrou et al., 2013; Weselek et al., 2021) and multi-model ensembles (Jägermeyr et al., 2021) should be used to confirm the results. (ii) The microclimate model is empirical: the canopy-cooling coefficient  $\kappa$  strongly affects yield retention (Section 3.5) and needs field calibration for trackers. (iii) Warming was applied as uniform temperature shifts with fixed relative humidity, precipitation and irradiance, not as bias-adjusted climate-model projections, and cultivars and sowing dates were not adapted. (iv) Fields are either rain-fed or fully irrigated, and irrigation water savings under the panels are not credited. (v) SALS meets the yield floor on average but not in every season. A chance-constrained or risk-averse objective, e.g. a conditional value-at-risk of the retention, is needed where the floor is enforced every year. (vi) The upscaled potentials are technical: grid capacity, storage, the value of midday

412 electricity, the cost of elevated structures and farmers’ acceptance (Tromms-  
413 dorff et al., 2021) will determine the realisable share.

## 414 5. Conclusions

415 We have presented SALS, a learning algorithm that designs and controls  
416 agrivoltaic systems end-to-end through a differentiable simulator of trackers,  
417 PV output, crop light, microclimate, soil water and crop growth. Trained on  
418 400 cropland sites of the world’s five major staple crops and four warming  
419 levels, SALS produced 72–85% of the electricity of a conventional solar plant  
420 while keeping the mean crop yield at 90% of the open field. This raised  
421 the land-equivalent ratio from 1.2 for static designs and 1.4 for expert light-  
422 sharing rules to 1.6–1.75 on geographically held-out sites and years. The  
423 learned policy shares light during canopy development and shades the crop  
424 during heat, and its advantage grows with warming. Scaled to 5% of the  
425 harvested area of the five crops, learned agrivoltaics would have a technical  
426 potential comparable to current global electricity generation. Future work  
427 should add chance constraints for year-by-year food security, crop-specific  
428 shade physiology and field validation of the learned control on commercial  
429 tracker systems.

## 430 CRediT authorship contribution statement

431 **Quang-Vinh Dang:** Conceptualization, Methodology, Software, Formal  
432 analysis, Investigation, Data curation, Visualization, Writing – original draft,  
433 Writing – review & editing.

## 434 Declaration of competing interest

435 The author has no known competing financial interests or personal re-  
436 lationships that could have appeared to influence the work reported in this  
437 paper.

## 438 Data and code availability

439 All data used are public: SPAM 2010 v2r0 harvested areas (Yu et al.,  
440 2020), the GGCM Phase 3 crop calendar (Jägermeyr et al., 2021), NASA  
441 POWER daily weather (Sparks, 2018) and the SIMPLE crop parameters

(Zhao et al., 2019). The code of the simulator, of SALS and of all experiments, together with the site list, is available from the author’s public repository (<https://github.com/vinhqdang/Artificial-Intelligence-in-Agriculture>).

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## Appendix A. Statistical tests

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Table 2: Held-out performance (test sites, seasons 2015–2019). GCR: mean ground coverage ratio;  $E$ : electricity ( $\text{MWh ha}^{-1} \text{yr}^{-1}$ );  $e$ : electricity relative to a conventional PV plant;  $r$ : mean yield retention; compliance: share of seasons with  $r \geq 0.9$ ; LER =  $r + e$  with bootstrap 95% CI over sites. Oracle-GCR baselines choose the GCR with knowledge of the test weather; the open-loop optimisation also knows the daily weather of each test season.

| Method                 | GCR  | $E$         | $e$  | $r$   | Compliance (%) | LER [95% CI]      |
|------------------------|------|-------------|------|-------|----------------|-------------------|
| <i>Baseline</i>        |      |             |      |       |                |                   |
| AV static (rule)       | 0.10 | 361         | 0.26 | 0.920 | 60.3           | 1.18 [1.16, 1.20] |
| AV static (oracle GCR) | 0.10 | 377         | 0.27 | 0.935 | 85.2           | 1.20 [1.16, 1.25] |
| Seasonal sharing       | 0.26 | 629         | 0.45 | 0.921 | 82.2           | 1.37 [1.33, 1.41] |
| Stress rule            | 0.23 | 707         | 0.50 | 0.922 | 80.9           | 1.42 [1.37, 1.48] |
| SALS (ours)            | 0.34 | <b>1019</b> | 0.72 | 0.909 | 49.5           | 1.63 [1.57, 1.70] |
| Open-loop (foresight)  | 0.30 | 916         | 0.65 | 0.901 | 28.6           | 1.55 [1.49, 1.61] |
| <i>+1.5 °C</i>         |      |             |      |       |                |                   |
| AV static (rule)       | 0.10 | 355         | 0.26 | 0.924 | 66.7           | 1.18 [1.16, 1.20] |
| AV static (oracle GCR) | 0.11 | 385         | 0.28 | 0.931 | 83.5           | 1.21 [1.17, 1.25] |
| Seasonal sharing       | 0.28 | 647         | 0.47 | 0.918 | 79.2           | 1.38 [1.34, 1.43] |
| Stress rule            | 0.23 | 722         | 0.52 | 0.919 | 77.7           | 1.44 [1.39, 1.49] |
| SALS (ours)            | 0.36 | <b>1079</b> | 0.77 | 0.901 | 46.8           | 1.67 [1.61, 1.74] |
| <i>+2 °C</i>           |      |             |      |       |                |                   |
| AV static (rule)       | 0.10 | 354         | 0.26 | 0.926 | 69.1           | 1.18 [1.16, 1.20] |
| AV static (oracle GCR) | 0.11 | 393         | 0.29 | 0.929 | 83.5           | 1.22 [1.17, 1.26] |
| Seasonal sharing       | 0.28 | 659         | 0.48 | 0.917 | 78.9           | 1.39 [1.35, 1.44] |
| Stress rule            | 0.23 | 728         | 0.53 | 0.918 | 77.2           | 1.44 [1.39, 1.50] |
| SALS (ours)            | 0.37 | <b>1106</b> | 0.79 | 0.898 | 46.3           | 1.69 [1.62, 1.76] |
| <i>+3 °C</i>           |      |             |      |       |                |                   |
| AV static (rule)       | 0.10 | 353         | 0.26 | 0.933 | 74.2           | 1.19 [1.17, 1.21] |
| AV static (oracle GCR) | 0.12 | 442         | 0.32 | 0.927 | 80.1           | 1.25 [1.20, 1.30] |
| Seasonal sharing       | 0.30 | 694         | 0.50 | 0.920 | 78.1           | 1.42 [1.38, 1.47] |
| Stress rule            | 0.24 | 762         | 0.55 | 0.919 | 77.7           | 1.47 [1.42, 1.53] |
| SALS (ours)            | 0.39 | <b>1173</b> | 0.85 | 0.901 | 45.1           | 1.75 [1.68, 1.82] |
| Open-loop (foresight)  | 0.34 | 1028        | 0.74 | 0.901 | 25.8           | 1.65 [1.58, 1.71] |

Table A.3: Paired differences between SALS and each baseline on the held-out test seasons (SALS minus baseline) and two-sided Wilcoxon signed-rank  $p$ -values.

| Climate  | Baseline               | $\Delta \text{LER}$ | $p$         | $\Delta e$ | $p$         | $n$ |
|----------|------------------------|---------------------|-------------|------------|-------------|-----|
| Baseline | AV static (rule)       | +0.452              | $< 10^{-4}$ | +0.463     | $< 10^{-4}$ | 461 |
| Baseline | AV static (oracle GCR) | +0.428              | $< 10^{-4}$ | +0.454     | $< 10^{-4}$ | 461 |
| Baseline | Seasonal sharing       | +0.264              | $< 10^{-4}$ | +0.276     | $< 10^{-4}$ | 461 |
| Baseline | Stress rule            | +0.207              | $< 10^{-4}$ | +0.220     | $< 10^{-4}$ | 461 |
| Baseline | Open-loop (foresight)  | +0.081              | $< 10^{-4}$ | +0.072     | 0.0012      | 461 |
| +1.5 °C  | AV static (rule)       | +0.493              | $< 10^{-4}$ | +0.515     | $< 10^{-4}$ | 462 |
| +1.5 °C  | AV static (oracle GCR) | +0.463              | $< 10^{-4}$ | +0.493     | $< 10^{-4}$ | 462 |
| +1.5 °C  | Seasonal sharing       | +0.290              | $< 10^{-4}$ | +0.306     | $< 10^{-4}$ | 462 |
| +1.5 °C  | Stress rule            | +0.235              | $< 10^{-4}$ | +0.252     | $< 10^{-4}$ | 462 |
| +2 °C    | AV static (rule)       | +0.510              | $< 10^{-4}$ | +0.537     | $< 10^{-4}$ | 460 |
| +2 °C    | AV static (oracle GCR) | +0.477              | $< 10^{-4}$ | +0.508     | $< 10^{-4}$ | 460 |
| +2 °C    | Seasonal sharing       | +0.299              | $< 10^{-4}$ | +0.318     | $< 10^{-4}$ | 460 |
| +2 °C    | Stress rule            | +0.249              | $< 10^{-4}$ | +0.269     | $< 10^{-4}$ | 460 |
| +3 °C    | AV static (rule)       | +0.558              | $< 10^{-4}$ | +0.590     | $< 10^{-4}$ | 457 |
| +3 °C    | AV static (oracle GCR) | +0.498              | $< 10^{-4}$ | +0.524     | $< 10^{-4}$ | 457 |
| +3 °C    | Seasonal sharing       | +0.324              | $< 10^{-4}$ | +0.342     | $< 10^{-4}$ | 457 |
| +3 °C    | Stress rule            | +0.276              | $< 10^{-4}$ | +0.295     | $< 10^{-4}$ | 457 |
| +3 °C    | Open-loop (foresight)  | +0.102              | $< 10^{-4}$ | +0.101     | $< 10^{-4}$ | 457 |